

The Biodiversity of *Dharmapur* Sal (*Shorea robusta*) Forests in Northwestern Bangladesh: Past and Present

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Abstract

The overarching objective of the study was to assess the change in biodiversity of the unexplored Northwestern Sal forests of Bangladesh. To do so, data was collected from the *Dharmapur* Sal forests, the last abode of the natural ecosystem in that region. Comparing with the historical data, it was revealed that 09 tree species, 08 shrub species, and 03 climber species are missing currently. Due to the disappearance of associate tree species, the forest has become pure stands representing only abundant Sal species. Sal can withstand an adverse environment, but the undergrowth including associate tree species cannot regenerate under severe anthropogenic disturbances. The archival data showed that once, the forest was extremely rich in faunal diversity with Indian Rhinoceros, Asian black bear, Barking deer, Asian elephants, Leopard, Blue bull, Spotted deer, White-rumped vulture, Red narrow-mouthed frog, Sambar deer, Capped langur, Leopard cat, Jungle cat, Small Indian civet, Indian flying fox, Russel's viper, and Rhesus macaques, but they became extinct locally over the time. In this day and age, the amphibian, reptile, and mammal communities are comprised of 09, 23, and 08 species respectively. The conservation strategies for the future were developed spotlighting the reforestation and *Status quo*. Recommendations were made to address the inherent problems that emerged from the society.

Keywords: Sal Forest, extinction, conservation, flora, fauna

Introduction

Until the beginning of the 20th century, the Sal (*Shorea robusta*) existed as a continuous belt with rich floral and faunal diversity (Rahman et al. 2009). Over time, the gradual increase of populations put horrendous pressure on this ecosystem by degrading the natural habitats badly. In addition, the Sal Forests of Bangladesh were severely damaged by the destructive anthropogenic activities with merciless overexploitation of forest resources (Rahman 2021). Together with this, the major portion of the Sal Forests was converted into exotic plantation forests, rubber gardens, agricultural and horticultural crop fields, and human settlements (Rahman et al. 2007; Rahman 2024a).

In Bangladesh, Sal forests are currently distributed sporadically in different geographical locations (Alam *et al.* 2008). The major portion of Sal forests in Bangladesh is located in the central parts notably in the Mymensingh, Tangail, and Gazipur districts. The north-western region and Comilla district have remnants of natural forests at present (Alam *et al.* 2008). Islam et al. (2011) reported that massive deforestation occurred over the last few decades in the northwestern Dinajpur district. Sharmi *et al.* (2021) revealed that the small population size of Sal in the *Dinajpur* forest might be related to its low genetic diversity.

Most of the Sal-centric studies in Bangladesh focused on the Madhupur Sal forests of central Mymensingh and Tangail districts (Alam *et al.* 2008). However, the northwestern Sal forests were unable to draw the attention of the researchers (Sharmi *et al.* 2021). Even a complete checklist of existing flora and fauna could be hardly found. Hence, the study aims to explore this unrepresented natural ecosystem. A complete list of available flora and fauna will the policymakers to take necessary steps for the conservation. The study also strived to compare the biodiversity status between the past and present.

Methodology

Study area

The study area was located in the *Biral Upazila* of Dinajpur district (Figure 1). Locally the natural patches are known as *Dharmapur* Sal forests which cover an area of 34540 acres of land. This is the last abode of natural Sal forests in Northwestern Bangladesh. It is 8 km from *Biral Upazila* and adjacent to *Dharmapur Union Parishad*.

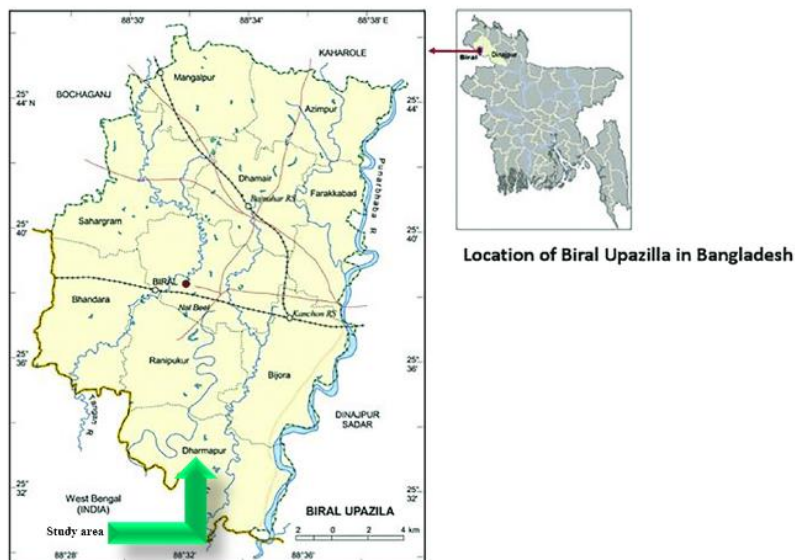


Figure 1: Location of the study area

Data Collection

The historical data of flora notably trees, shrubs, and climbers were collected from the local Forest Beat Office. Unfortunately, such data regarding wildlife were absent. The secondary data was collected from national newspapers, journals, books, and archives.

A total of four belt transects covering a total area were systematically surveyed on foot. In belt-transect sampling, the researchers accompanied by two local inhabitants slowly moved and recorded the flora and fauna. Photographic and video records were kept. The night survey was conducted in early and late nights with the help of high-power torches and headlamps to spot the nocturnal and cryptic animal. A total number of 100 local people were personally interviewed to know the past and present abundance of flora and wildlife. A focus group discussion was done incorporating the respondents from various stakeholders to find out proper conservation strategies.

Results and Discussions

Flora

Tree

In the past, the dominant species Sal (*Shorea robusta*) had 23 abundant associate tree species. Currently, the forests have turned into a pure stand, where Sal is the only abundant species. Meanwhile, 09 abundant associate tree species have disappeared (Table 1). Among the available tree species, *Jika/Jiol* (*Lannea coromandelica*), *Bonjam* (*Ardisia solanacea*), and *Sonalu* (*Cassia fistula*) were occasionally distributed. Concurrently, 10 species were rarely recorded.

In another study, Jashimuddin *et al.* (1999) revealed that Sal was the main species in *Biral*, but a handful of associates enhanced its diversity. Sal can withstand an adverse environment in its

natural habitat and adapt to the changing environments. However, the associate species cannot regenerate and survive under severe anthropogenic disturbances (Rahman *et al.* 2009, Rahman 2010, Rahman *et al.* 2007; Rahman and Vacik 2009).

Table 1: Present status of the past abundant tree species

<i>SL</i>	<i>Local Name</i>	<i>Common Name</i>	<i>Botanical Name</i>	<i>Present status</i>
1	Sal	Sal	<i>Shorea robusta</i>	A
2	Jika/Jiol	Indian ash tree	<i>Lannea coromandelica</i>	O
3	Bela	Marking nut	<i>Setnecarpus anacaridum</i>	R
4	Kachchhara	Ceylon hydrolea	<i>Hydrolza zeylanica</i>	R
5	Pora Amloki	Tahitian gooseberry	<i>Phyllanthus acidus</i>	N
6	Chalta	Elephant Apple	<i>Dillenia indica</i>	R
7	Bahera	Beach-almond	<i>Terminalia belerica</i>	R
8	Shimul	Malabar silk-cotton	<i>Bombax ceiba</i>	R
9	Lakkha Pakur	Weeping fig	<i>Ficus benjamina</i>	R
10	Haritaki	Black Myrobalan	<i>Terminalia reticulata</i>	R
11	Furush	Crepe Myrtle	<i>Lagtrstroemia indica</i>	R
12	Pitali	False White Teak	<i>Trewia nudiflora</i>	N
13	Palash	Bastard Teak	<i>Butea monospanna</i>	R
14	Baichi	Governor's plum	<i>Flacourtia indica</i>	N
15	Dumur	Hairy Fig	<i>Ficus hispida</i>	R
16	Bot	Indian Banyan	<i>Ficus benghalensis</i>	N
17	Jhula Bot	Golden Rumph's Fig	<i>Ficus rumphii</i>	R
18	Kharajora	Many-Flowered Litsea	<i>Litsea monopetala</i>	N
19	Mohua	Butter Tree	<i>Madhuca indica</i>	N
20	Jibon	Charcoal-tree	<i>Trema orientalis</i>	N
21	Bonjam	Coralberry	<i>Ardisia solanacea</i>	O
22	Lal Sonalu	Red Cassia	<i>Cassia roxburghii</i>	N
23	Sonalu	Golden shower	<i>Cassia fistula</i>	O
24	Golapi Sonalu	Pink Cassia	<i>Cassia javanica</i>	N

Shrub

In the past, 20 species of shrubs lived under the trees. They were distributed abundantly and regularly. Over time, 08 species became locally extinct (Table 2). Only 06 species namely *Bijrilata* (*Capparis zeylanica*), *Datmajani* (*Glycosmis pentaphylla*) *Sarpagandha* (*Rauwolfia serpentina*), *Kalkasunda* (*Senna occidentalis*), *Tit Begun* (*Solanum violaceum*), and *Anaigota* (*Ziziphus xylopyrus*) were occasionally found. The rest have become rare species.

Like tree associates, the shrub cannot cope with human disturbances (Rahman *et al.* 2009). They are highly extracted for medicinal values and fuelwood. As a result, the shrub community disappears from the natural forests gradually.

Table 2: Present status of the past abundant shrub species

<i>SL</i>	<i>Local Name</i>	<i>Common Name</i>	<i>Botanical Name</i>	<i>Present status</i>
1	<i>Bijrilata</i>	Ceylon Caper	<i>Capparis zeylanica</i>	O
2	<i>Jhijhi Jam</i>	Thompson's Marlberry	<i>Ardisia thomsonii</i>	N
3	<i>Ketoki</i>	Crepe Ginger	<i>Cheilocostus speciosus</i>	N
4	<i>Bon Motmotia</i>	Devil Weed	<i>Chromolaena odorata</i>	N
5	<i>Makunda</i>	Painted Leaf Croton	<i>Croton persimilis</i>	R

6	<i>Gol Mohani</i>	Redberry	<i>Deeringia amaranthoides</i>	R
7	<i>Modhupai Dhaira</i>	Honey Suckled Mistletoe	<i>Dendrophthoe falcata</i>	R
8	<i>Datmajani</i>	Gin berry	<i>Glycosmis pentaphylla</i>	O
9	<i>Kamala</i>	Red Kamala	<i>Mallotus philippensis</i>	N
10	<i>Akush</i>	Climbing Mallotus	<i>Mallotus repandus</i>	R
11	<i>Langka Jaba</i>	Sleeping hibiscus	<i>Malvaviscus glabrecens</i>	N
12	<i>Panjuli</i>	Black-Honey Shrub	<i>Phyllanthus reticulatus</i>	R
13	<i>Chitra</i>	Ceylon leadwort	<i>Plumbago zeylanica</i>	N
14	<i>Sarpagandha</i>	Devil pepper	<i>Rauwolfia serpentina</i>	O
15	<i>Kalkasunda</i>	Septicweed	<i>Senna occidentalis</i>	O
16	<i>Berela</i>	Common Wireweed	<i>Sida acuta</i>	R
17	<i>Lal Berela</i>	Jelly leaf	<i>Sida rhombifolia</i>	N
18	<i>Tit Begun</i>	Poison Berry	<i>Solanum violaceum</i>	O
19	<i>Chandaprabha</i>	Yellow Elder	<i>Tecoma stans</i>	N
20	<i>Anaigota</i>	Kath Ber	<i>Ziziphus xylopyrus</i>	O

Climbers

The historical data showed that *Dharmapur* Sal forests inhabited 13 abundant species of climbers in the past (Table 3). Currently, 03 species namely *Podolata Kachu* (*Syngonium podophyllum*), *Bilari* (*Mukia maderaspatana*), and *Anguri* (*Cissus verticillata*) are missing. It revealed that *Makal* (*Trichosanthes tricuspidata*), *Akundi* (*Stephania japonica*), *Kumarilata* (*Smilax ovalifolia*), *Shial Leza* (*Rhynchosyilis retusa*), *Jhumka Lata* (*Passiflora foetida*), *Swarna Lata* (*Cuscuta reflexa*), and *Ishwarmul* (*Aristolochia lanceolata*) were sporadically distributed across the forests. Three species: *Gajapipul* (*Scindapsus officinalis*), *Ban Alu* (*Dioscorea bulbifera*), and *Pani Alu* (*Dioscorea oppositifolia*) were rarely recorded.

The climbers cannot regenerate in a disturbed habitat like Sal associate tree species and climbers (Rahman *et al.* 2009). The community uses climbers as vegetables and medicines. The species richness of climbers is related to the species richness of trees (Rahman *et al.* 2010).

Table 3: Present status of the past abundant climber species

SL	Local Name	Common Name	Botanical Name	Present status
1	<i>Makal</i>	Red ball Snake gourd	<i>Trichosanthes tricuspidata</i>	O
2	<i>Podolata Kachu</i>	Arrowhead vine	<i>Syngonium podophyllum</i>	N
3	<i>Akundi</i>	Snake vine	<i>Stephania japonica</i>	O
4	<i>Kumarilata</i>	Catbriers	<i>Smilax ovalifolia</i>	O
5	<i>Gajapipul</i>	Pipul	<i>Scindapsus officinalis</i>	R
6	<i>Shial Leza</i>	Foxtail orchid	<i>Rhynchosyilis retusa</i>	O
7	<i>Jhumka Lata</i>	Stinking passionflower	<i>Passiflora foetida</i>	O
8	<i>Bilari</i>	Rough Bryony	<i>Mukia maderaspatana</i>	N
9	<i>Ban Alu</i>	Air potato	<i>Dioscorea bulbifera</i>	R
10	<i>Pani Alu</i>	Chinese yam	<i>Dioscorea oppositifolia</i>	R
11	<i>Swarna Lata</i>	Giant dodder	<i>Cuscuta reflexa</i>	O
12	<i>Anguri</i>	Princess Vine	<i>Cissus verticillata</i>	N
13	<i>Ishwarmul</i>	Indian Birthwort	<i>Aristolochia lanceolata</i>	O

Fauna

Extinct species

The archival data showed that once, the forest was extremely rich in faunal diversity with Indian Rhinoceros, Asian black bear, Barking Deer, Asian elephants, Leopard, Blue Bull, Spotted Deer, White-rumped Vulture, Red Narrow Mouthed Frog, Sambar deer, Capped Langur, Leopard cat, Jungle cat, Small Indian civet, Indian Flying Fox, Russel's Viper, and Rhesus macaques, but they became extinct locally over the time (Table 4).

The natural forests turned into small remnants, which are also highly fragmented. The introduction of exotic plantations has altered the environment. The insufficient tree density and forest coverage do not support a home range for the larger animals in heavily fragmented forests. Species also tend to suffer from genetic drift and inbreeding in fragmented forests (Rahman 2023, a). Fragment edges are also unfriendly to them, as the community often prosecutes them. Rahman (2024a) reported that a lack of biological corridors, buffer zones, and forest matrix forced Blue Bull to be extinct from the Sal forests of Bangladesh. The destruction of vegetation coverage in the upper part of the Padma Basin facilitated the movement of Russell's vipers downstream (Rahman 2024). The vulture population in Bangladesh faces an onslaught of threats from veterinary painkilling drugs like Diclofenac and Ketoprofen (Rahman 2023b).

Table 4: A list of extinct animals

19 th century		20 th century		21 st century	
Common Name	Scientific Name	Common Name	Scientific Name	Common Name	Scientific Name
Indian Rhinoceros	<i>Rhinoceros unicornis</i>	Asian Elephant	<i>Elephas maximus</i>	Capped Langur	<i>Trachypithecus pileatus</i>
Asian black bear	<i>Ursus thibetanus</i>	Leopard	<i>Panthera pardus</i>	Leopard cat	<i>Prionailurus bengalensis</i>
Barking Deer	<i>Muntiacus muntjak</i>	Blue Bull	<i>Boselaphus tragocamelus</i>	Jungle cat	<i>Felis chaus</i>
		Spotted Deer	<i>Axis axis</i>	Small Indian civet	<i>Viverricula indica</i>
		White-rumped Vulture	<i>Gyps bengalensis</i>	Indian Flying Fox	<i>Pteropus giganteus</i>
		Sambar deer	<i>Rusa unicolor</i>	Russel's Viper	<i>Daboia Russelii</i>
		Red Narrow Mouthed Frog	<i>Microhyla rubra</i>	Rhesus macaques	<i>Macaca mulatta</i>

Amphibians

The amphibians represent only 10 species of toads in this forest (Table 5). Only the common toad (*Duttaphrynus melanostictus*) was frequently found, whereas the Common Tree Frog (*Polypedates leucomystax*) was occasionally distributed. The rest were rarely spotted.

Table 5: A list of currently found amphibians

SL	Common Name	Scientific Name	Distribution
1	Common Toad	<i>Duttaphrynus melanostictus</i>	A

2	Skipper Frog	<i>Euphlyctis cyanophlyctis</i>	R
3	Kalasgram Skipper Frog	<i>Euphlyctis kalasgramensis</i>	R
4	Nepal Wart Frog	<i>Fejervarya nipalensis</i>	R
5	Pierre's Cricket Frog	<i>Fejervarya pierrei</i>	R
6	Terai Wart Frog	<i>Fejervarya teraiensis</i>	R
7	Jerdons Bullfrog	<i>Hoplobatrachus crassus</i>	R
8	Indian Bullfrog	<i>Hoplobatrachus tigerinus</i>	R
9	Narrow-mouthed Frog	<i>Microhyla sp</i>	R
10	Common Tree Frog	<i>Polypedates leucomystax</i>	O

Reptiles

The reptile community is represented by 01 lizard, 07 snakes, and 01 monitor species (Table 6). Common Garden Lizards (*Calotes versicolor*) and Common Vine Snakes (*Ahaetulla nasuta*) are common throughout the forests.

Table 6: A list of currently found reptiles

SL	Common Name	Scientific Name	Distribution
1	Common Garden Lizard	<i>Calotes versicolor</i>	A
2	Common Vine Snake	<i>Ahaetulla nasuta</i>	A
3	Common Bronze-back	<i>Dendrelaphis pictus</i>	R
4	Indian Rat Snake	<i>Ptyas mucosa</i>	O
5	Checkered Keelback	<i>Xenochrophis piscator</i>	R
6	Binocellate Cobra	<i>Naja naja</i>	R
7	Tokay Gecko	<i>Gekko gekko</i>	R
8	Diard's Blindsnake	<i>Argyrophis diardii</i>	R
9	Bengal Monitor	<i>Varanus bengalensis</i>	O

Birds

A total number of 23 bird species were counted (Table 7). Out of them, 10 species were occasionally found and the rest were rarely spotted.

Table 7: A list of currently found birds

SL	Common Name	Scientific Name	Distribution
1	Grey-headed Fish-eagle	<i>Ichthyophaga ichthyaetus</i>	R
2	Common Iora	<i>Aegithina tiphia</i>	R
3	Little Egret	<i>Egretta garzetta</i>	R
4	Small Minivet	<i>Pericrocotus cinnamomeus</i>	O
5	Asian Openbill	<i>Anastomus oscitans</i>	O
6	Rock Dove	<i>Columba livia</i>	R
7	Eastern Spotted Dove	<i>Spilopelia chinensis</i>	O
8	Eurasian Collared Dove	<i>Streptopelia decaocto</i>	R
9	Yellow Footed Green Pigeon	<i>Treron phoenicopterus</i>	R
10	Jungle Crow	<i>Corvus leuallantii</i>	O
11	Common Hawk-Cuckoo	<i>Hierococcyx varius</i>	R
12	Hair-crested Drongo	<i>Dicrurus hottentottus</i>	R
13	Black Drongo	<i>Dicrurus leucophaeus</i>	R
14	White Wagtail	<i>Motacilla alba</i>	O
15	Oriental Magpie-robin	<i>Copsychus saularis</i>	O

16	Great Tit	<i>Parus major</i>	R
17	House Sparrow	<i>Passer domesticus</i>	O
18	Little Cormorant	<i>Microcarbo niger</i>	R
19	Fulvous-breasted Woodpecker	<i>Dendrocopos macei</i>	O
20	Streak-throated Woodpecker	<i>Picus xanthopygaeus</i>	O
21	Common Myna	<i>Acridotheres tristis</i>	O
22	Common Tailorbird	<i>Orthotomus sutorius</i>	R
23	Jungle Babbler	<i>Turdoides striata</i>	R

Mammals

A total number of 08 mammals were spotted in this forest ecosystem. The Bengal Fox, Rabbit, House Mouse, and Five-striped Palm Squirrel were occasionally spotted, while the Golden Jackal, Fishing Cat, Lesser Bandicoot Rat, and Large Bandicoot Rat were distributed rarely (Table 8). Haque *et al* (2020) reported that hunting by ethnic *Santal* people badly impacted the mammalian diversity in the natural forests of the *Dinajpur* district.

Table 8: A list of currently found amphibians

<i>SL</i>	<i>Common Name</i>	<i>Scientific Name</i>	<i>Distribution</i>
1	Golden Jackal	<i>Canis aureus</i>	R
2	Bengal Fox	<i>Vulpes bengalensis</i>	O
3	Rabbit	<i>Oryctolagus cuniculus</i>	O
4	Fishing Cat	<i>Prionailurus viverrinus</i>	R
5	Lesser Bandicoot Rat	<i>Bandicota bengalensis</i>	R
6	Large Bandicoot Rat	<i>Bandicota indica</i>	R
7	House Mouse	<i>Mus musculus</i>	O
8	Five-striped Palm Squirrel	<i>Funambulus pennantii</i>	O

Conservation for the future

Based on the respondents' perceptions, conservation strategies were developed. Without the proper enforcement of laws and strictly controlling further logging of Sal and its associates, it is impossible to maintain this forest ecosystem functional. Controlling the rampant encroachment of forest land is the prime condition before initiating any conservation activity. It is important to demark the forest lands into core, buffer, and peripheral zones. Restoration of forest lands through reforestation by early successional Sal species can expand the natural habitats. Gap-filling between the fragments will initiate biological corridors, buffer zones, and forest matrix for the wildlife. Natural regeneration notably natural succession in the adjacent areas of Sal patches will restore the natural vegetation. The exotic plantation should be stopped as it alters the natural habitats and provides mayhem for the wildlife (Rahman 2023c, d). It is important to preserve a portion of deadwood for the birds. Rahman (2023e) revealed that without the presence of standing deadwood, it is difficult for woodpeckers to survive. Under the social safety net programme, alternative income-generating activities and the supply of alternative fuel sources for the forest people will reduce the pressure on natural resources. A conservation campaign will build awareness among the people.

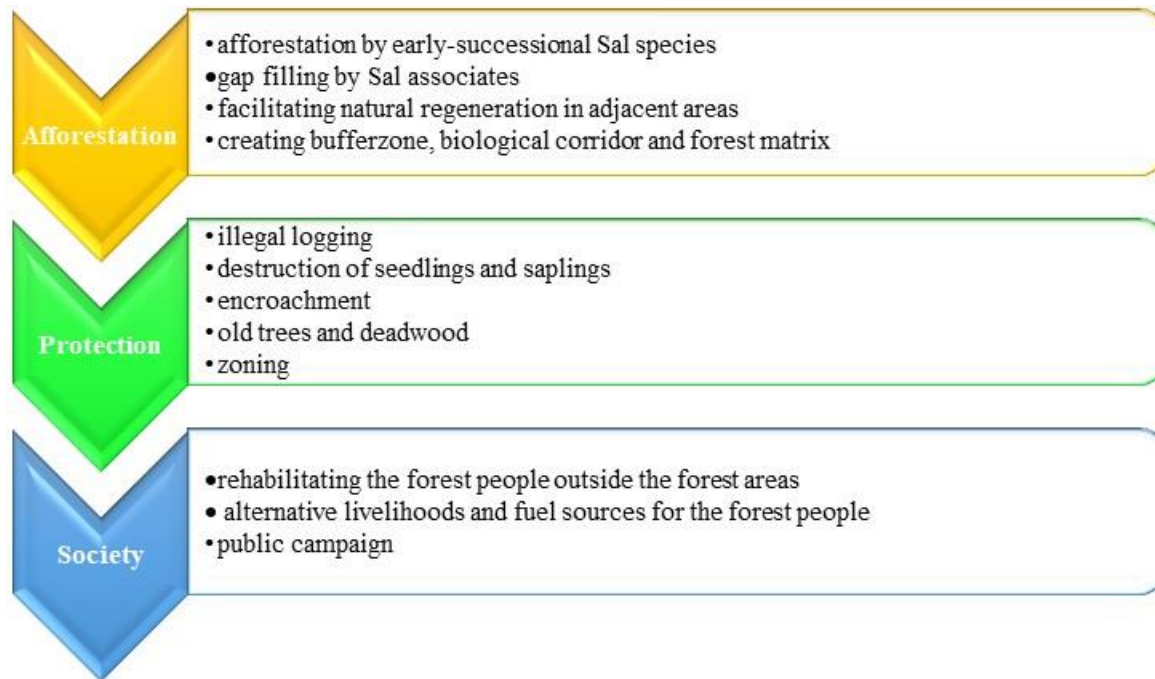


Figure 2: Conservation strategies for this forest ecosystem

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